

Nicholas Mesa-Cucalon

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EXPERIENCE

- **Carnegie Mellon University** Pittsburgh, PA
Student Researcher Aug 2024 – Present
 - **Human Sensing Lab - Lead Author (In-Progress):** Designing a quantitative framework to measure modality bias in Video-Language Models, managing the full experimental pipeline on an NVIDIA A40 GPU.
 - **InfiniAI Lab - VLM Optimization:** Performed literature review on state-of-the-art KV-cache optimization to implement a Heavy Hitter Oracle KV-Cache in LLaVA-1.5 7B. Matched vanilla performance on TextVQA (61.24%) using only 25% of the original KV-Cache size while maintaining inference speed.
- **REEP Equity** San Antonio, TX
AI Software Intern May 2025 – Aug 2025
 - **Prototype Development:** Built an end-to-end app with a web frontend and Flask/Python backend. Processed analyst CSV data and leveraged in-context learning with a LLaMA-3.2 3B model to generate summaries.
 - **User Testing and Validation:** Conducted A/B testing with clients comparing AI-generated and human-written summaries, demonstrating no significant preference and validating the prototype's effectiveness.
- **Voage** Pittsburgh, PA
Robotics and Perception Intern May 2023 – Aug 2023
 - **Robot Calibration:** Led development and deployment of production-grade Delta Robot Calibration system with an 8-member team. Designed and documented a Hand-Eye Calibration solution for xArm7 robotic arms integrating Intel RealSense and Zivid cameras with OpenCV.
 - **Point Cloud Registration Research:** Conducted a literature review of 23 academic papers on Point Cloud Registration to synthesize key methods for development of core product.

PROJECTS

- **Dendritic Optimization for TD3 Reinforcement Learning:** Won 3rd place in a Hackathon with 693 participants. Reimplemented TD3 on DMControl with PerforatedAI's Artificial Dendrite Networks. Integrated Lower Confidence Bound validation to reduce return variance by 70%. Achieved 13.1% improvement in mean return over baseline.
- **Multimodal Meme Caption Generation:** Developed and implemented a model to leverage hybrid and late fusion of text and image data to generate captions for memes on the MemeCap dataset. Model achieved a BLEU-4 F1 Score of 9.0 and ROUGE-L F1 Score of 25.9, comparable to state-of-the-art models.
- **Denoising Diffusion Probabilistic Models Implementation:** Wrote code implementing forward process, reverse process and forward sampling for DDPM. Trained on cat images of the AFHQ dataset with a fixed compute of a single A100 GPU. Model was able to produce realistic cat images after training for 10,000 steps in 20 minutes.

EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA
B.S. in Electrical and Computer Engineering; Minor in Machine Learning; GPA: 3.76 Aug 2021 – May 2025
 - **Relevant Courses:** Computer Systems & Hardware-Software Interface, Distributed Systems, Machine Learning Systems, Bayesian Methods in Machine Learning, Generative AI
- **Carnegie Mellon University** Pittsburgh, PA
M.S. in Electrical and Computer Engineering; Concentration in AI/ML Systems; GPA: 4.0 Aug 2025 - May 2026
 - **Relevant Courses:** Multimodal Machine Learning, Deep Reinforcement Learning, Speech Recognition and Understanding, Deep Learning for Computer Vision, ML for Cybersecurity, How to Write Fast Code II

SKILLS

- **Programming Languages:** Python, C, C++, Golang, Swift
- **Programming Frameworks:** PyTorch, OpenCV, Huggingface, Pandas, NumPy, LangGraph, LangChain
- **Technologies:** Deep Learning, Computer Vision, Natural Language Processing, Weights and Biases, Git, Linux, AWS
- **Languages:** English (Native), Spanish (Native)